



Neural mechanisms of sexual decision-making in women with alcohol use disorder

Kess L. Folco¹ · Daniel J. Fridberg² · Lindsay R. Arcurio¹ · Peter R. Finn¹ · Julia R. Heiman¹ · Thomas W. James¹

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Abstract

Rationale The co-occurrence of alcohol consumption and sexual activity is associated with increased risk for sexual assault, sexually transmitted disease, and unplanned pregnancy among young adult women with alcohol use disorder (AUD). There is considerable previous work demonstrating neural reactivity to alcohol cues in AUD. Because alcohol consumption and sexual behavior are both rewarding and tend to co-occur, sexual cues may produce similar neural reactivity in women with AUD, possibly indicating a shared mechanism underlying reactivity to both types of cues. Alternatively, reactivity to alcohol versus sexual cues may be distinct, suggesting domain-specific mechanisms.

Objectives We investigated whether the decision vulnerabilities in AUD women regarding sexual activity were related to differences in brain activation compared to control women.

Methods Women with ($n = 15$) and without ($n = 16$) AUD completed a hypothetical decision-making task during fMRI that presented low- or high-risk scenarios involving visual sexual, appetitive, and neutral cues.

Results Results showed that sexual cues were more often endorsed by women with AUD compared to controls and elicited differential brain activation patterns in frontal, visual, and reward regions. During high-risk decisions, women with AUD failed to downregulate activation, causing hyperactivation compared to controls.

Conclusions Visual sexual cues produced reactivity like that previously demonstrated for alcohol cues, suggesting a shared or domain-general mechanism for alcohol and sexual cue reactivity in women with AUD. Riskier sexual decisions in women with AUD may be a consequence of repeatedly pairing alcohol use and sexual activity, a characteristic behavior of this population.

Keywords Female sexual risk · fMRI · Alcohol use disorder reward networks · Incentive sensitization · Decision-making · Alcohol use disorder · Cue reactivity · Co-sensitization

Introduction

Alcohol consumption has a situational influence on sexual decision-making, especially in young adults (Norris et al. 2009; Testa et al. 2000; Purdie et al. 2011; Stoner et al. 2008), and intoxication contributes to the high co-occurrence of alcohol use and risky sexual behavior (Davis et al. 2007; Dir et al. 2017; Zule et al. 2018; Abbey 2011; O'Hare 1998). Alcohol intoxication has been associated with increased sexual arousal,

increased engagement in risky sexual activities, and increased risk of sexual assault (Ullman 2003; Gidycz et al. 2006; Koss et al. 1987; Dudley 2005; Parks et al. 2008). Alcohol itself, however, is unlikely to be the only variable contributing to these associations. Individuals who tend to drink more also show general disinhibition with impulsive decision-making in the laboratory and a high prevalence of externalizing disorders (Bailey et al. 2018; Finn 2002; Bobova et al. 2009). **Although intoxication leads to riskier decisions in all individuals, impulsive individuals and chronic alcohol users make riskier decisions independent of intoxication** (Robinson and Berridge 1993, 2008; Yalachkov et al. 2010; Bailey et al. 2018; Endres et al. 2014; Bechara 2005). When alcohol use and sexual activity are combined, women are at greater risk than men for negative outcomes, including contracting a sexually transmitted infection, unplanned pregnancy, and experiencing sexual assault (Dir et al. 2017; Zule et al. 2018; Abbey 2011; O'Hare

✉ Thomas W. James
thwjames@indiana.edu

¹ Department of Psychological and Brain Sciences, Indiana University, 1101 E 10th St, Bloomington, IN 47405, USA

² Department of Psychiatry and Behavioral Neuroscience, The University of Chicago, Chicago, IL, USA

1998). Despite these strong associations, the influence of chronic alcohol use in young adult women on sexual decision-making and brain activity has not been fully examined.

Impulsivity, externalizing disorders, and conduct disorders are highly correlated with decision variables related to drinking and sexual risk behaviors (Wilson and Vassileva 2016; Kareken et al. 2010; Dick et al. 2006). Furthermore, there is evidence that poor decision-making is correlated with developing alcohol use disorder (AUD) and this association may be mediated by genetic factors (Villafuerte et al. 2013; Chassin et al. 1999). For instance, individuals with family histories of alcohol misuse and individuals with the **GABRA2 gene** are more likely to develop AUD, make impulsive decisions on laboratory tests (e.g., delayed discounting), and have high comorbidity with conduct disorders (Kareken et al. 2010; Dick et al. 2006; Bailey et al. 2018; Endres et al. 2014; Arcurio et al. 2015; Czaplá et al. 2017). Thus, while alcohol intoxication may contribute **to riskier sexual decision-making while intoxicated, it may also amplify a dispositional tendency toward disinhibition in general.** This disinhibition may be intensified by repeated alcohol use and also may be exacerbated particularly for behaviors that are commonly associated with alcohol use in young adults, such as sexual activity.

Functional neuroimaging is a valuable tool for examining the neural correlates of the disinhibition exhibited by individuals with substance use disorder. Early models suggested a role of **hyperactivation in reward regions**, such as the striatum, with less-specific contributions from a variety of frontal cortical regions (Goldstein and Volkow 2002). Recent work has highlighted the roles of frontal regions that belong to different intrinsic brain networks, specifically the medial prefrontal cortex (mPFC) of the default mode network (DMN), the dorsal anterior cingulate cortex (dACC) of the salience network (SN), the middle frontal gyrus (MFG) of the central executive network (CEN), and the frontal pole of the (SN) (Goldstein and Volkow 2011; Jentsch and Taylor 1999; Grüsser et al. 2004). Differential recruitment of these regions may indicate a dysfunction in their interaction. The SN, CEN, and DMN show hyperactivation in individuals with AUD while making alcohol decisions (Arcurio et al. 2015). Critically, it has been suggested that the SN shuttles information efficiently and appropriately between the CEN and DMN, acting like a hub or network switch (Menon and Uddin 2010). These regions are implicated across different types of substance use, including in individuals with AUD (Leyton 2007; Koob and Le Moal 2005). **Importantly, most of these same regions are also implicated in sexual decision-making in non-clinical samples** (Baird et al. 2007; Afonso et al. 2007; Hyman et al. 2006). Finally, more recent inquiries have demonstrated a role for the sensory cortex, **especially the visual cortex,** in both general substance use and AUD (Yalachkov et al. 2010; Hanlon et al. 2014). Thus, frontal, visual, and reward brain regions from

among the SN, CEN, and DMN intrinsic brain networks are among the most affected in the AUD population (Yalachkov et al. 2010; Goldstein and Volkow 2011; Leyton 2007).

An outstanding question in the literature is whether young women with AUD make different sexual decisions than healthy control women and, if so, which brain networks are implicated in that difference. That is, does chronic alcohol misuse affect neural systems that process cues related to sexual decisions? An additional question is to what extent the **differences in brain activation with sexual cues in AUD women mimic those previously found with alcohol cues.** Here, we planned to address these questions by recruiting samples of women with and without AUD and asking them to complete a functional neuroimaging protocol while making hypothetical sexual decisions that varied along with different levels of risk (Arcurio et al. 2015; Nikoulina et al. 2020). In women with AUD, who have a history of pairing alcohol consumption and sexual activity, we expected to see a pattern of activation increase for sexual decisions in fronto-striatal systems and visual regions, similar to the pattern found previously for alcohol decisions (Arcurio et al. 2015; Yalachkov et al. 2010).

Methods and materials

Participants

Thirty-one female, heterosexual participants (16 controls, 15 AUD) between the ages of 18–38 were recruited as part of a larger study. A description of the recruitment strategy and detailed study inclusion/exclusion criteria are presented elsewhere (Arcurio et al. 2015). General inclusion criteria were regular 28–32-day menstrual cycles, not pregnant or using hormonal contraceptives within the last 3 months, no other current or past drug use except for occasional cannabis use (to allow for the high rates of co-occurrence between alcohol and cannabis use in this population; see Supplementary Information for criterion), no contraindications for MRI, not currently seeking treatment for alcohol misuse, no symptoms of psychosis or TBI, reported consuming at least one full drink of alcohol, and not currently abstaining from alcohol use. Additional inclusion criteria for the AUD group were (1) current DSM-IV alcohol dependence, (2) no current use of opiates, sedatives, or stimulants, and (3) no current DSM-IV cannabis dependence. Additional inclusion criteria for the control group were (1) no recreational drug use in the last 3 months, (2) no lifetime history of drug use (excluding cannabis), (3) fewer than 25 lifetime uses of cannabis, (4) not currently abstaining from alcohol, and (5) no current or past history of DSM-IV drug or alcohol abuse or dependence, see Table 2 for participant demographics and characterizations.

Test session exclusion criteria

Test sessions included the following exclusion criteria: (1) did not refrain from drinking alcohol and/or using any illicit psychoactive drug for a period of at least 24 h before testing, (2) did not refrain from sexual activity with a partner for 24 h prior to the test session, (3) did not refrain from eating within 4 h before the test session. At each test session, participants submitted to a breath alcohol test using an AlcoSensor IV (Intoximeter, Inc., St. Louis, MO) and a urine drug screen and answered questions about sexual activity with a partner within the past 24 h and food intake within the past 4 h. The drug screen was performed with a NEW CAD multi-drug screen test panel, which was a 5-panel screen for cocaine, methamphetamine, marijuana, opiates, and benzodiazepines. Participants were asked to reschedule the test session if their breath alcohol concentration was greater than 0.0%, if their urine sample was positive for any illicit drugs (including cannabis), or they did not meet our other test session requirements.

Alcohol use

In an interview, participants were asked if they regularly consumed alcohol or other drugs on each day of the week, and if yes, how much they usually consumed. Alcohol use was quantified as the sum of the usual amount of alcohol consumed for each day of the week, and the number of days per week where drinking usually occurred within the past 3 months. Drug use was quantified as the frequency of lifetime use for cannabis, stimulants, sedatives, and opiates. The Semi-Structured Assessment for the Genetics of Alcoholism (SSAGA; Bucholz et al. 1994) was used to determine whether participants satisfied DSM-IV (American Psychiatric Association 1994) diagnostic criteria for AD, cannabis dependence, and drug dependence. Alcohol and cannabis problem counts were also calculated from the SSAGA. Participants who reported the use of substances other than alcohol or cannabis were excluded; some cannabis use was permitted due to the high co-occurrence between alcohol and cannabis use in individuals with AUD (Finn et al. 2009).

Sexuality, mood, and impulsivity

Participants completed measures of mood, sexual behaviors, sexual arousal, and sexual decision-making. Specific measures are summarized in Table 1, see Table 2 for sample summary metrics. Mood was assessed using the Positive and Negative Affect Schedule (PANAS; Watson et al. 1988; $\alpha = .77$) and the Beck Depression Inventory-II (BDI; Beck et al. 1996; $\alpha = .88$). Recent sexual behavior was assessed using a brief 3-item questionnaire, which asked about the current sexual relationship(s), lifetime number of sexual partners,

and condom use. Each of the former items was assessed individually. Participants' experiences with and attitudes toward uncommitted sexual relations were measured using the 7-item Sociosexual Orientation Inventory (SOI; Simpson and Gangestad 1991; $\alpha = .82$). Overall sexual functioning was assessed using the 22-item Brief Index of Sexual Functioning for Women (BISF-W; Taylor et al. 1994; $\alpha = .87$) with the scoring system developed by Mazer et al. (2000). Individual differences in the propensity for sexual inhibition and excitation were measured using the 45-item Sexual Inhibition Scale/Sexual Excitation Scale (SIS/SES; Janssen et al. 2002), which consists of three subscales: Excitation (SES; $\alpha = .84$), Inhibition Due to the Threat of Performance Failure (SIS1; $\alpha = .75$), and Inhibition Due to Threat of Performance Consequences (SIS2; $\alpha = .70$). Participants' histories of unwanted sexual experiences due to verbal or physical coercion, or when under the influence of alcohol or drugs, were assessed with the 13 item Sexual Experiences Survey (Koss et al. 1987; $\alpha = .77$). Last, a general propensity toward risky behavior was assessed using the Evaluation of Risks Scale (EVAR) (Killgore et al. 2006; $\alpha = .70$).

Imaging procedure, parameters, and materials

Stimuli

There were four categories of cues: male faces, food, household/stationary items (plus alcoholic beverages, the data from which were presented previously (Arcurio et al. 2015), were not the focus of this article, and which were not of interest in any of the current analyses). Male faces formed the sexual decision cue category, food was the appetitive cue, and household items composed the neutral cue category. Pictures in each category were matched for arousal, valence, and desirability based upon normative data from a separate sample of university undergraduates as described previously (Arcurio et al. 2015). Risk information was used to create both a low- and high-risk context for each picture. The risk information consisted of the word "yes" or "no" and a single number. The low-risk context was always created with low-risk information, i.e., both parts of the risk information were low risk, whereas the high-risk context was always created with high-risk information.

Procedure

Participants were scheduled for two fMRI sessions following the initial interview session. As part of the larger project, each participant was scanned specifically at the follicular and luteal phases of their menstrual cycles with the order of the two sessions for each participant determined by which of the two phases was most imminent at the time of initial

Table 1 Sexuality and mood assessments, scale cited in online resources

	Items (components)	α	Score range	Description
Sexuality				
Sexual Experiences Survey (SES)	13	0.77	-	History of unwanted sexual experiences
Sociosexual Orientation Inventory (SOI)	7	0.82	-	Experiences and attitudes toward uncommitted sexual relationship(s)
Brief Index of Sexual Functioning for Women (BISF-W)	22 (64)	0.87	- 16:75	Overall sexual functioning
<i>Subscales</i>				
D1: Thoughts/Desires	2 (8)	-	0:12	-
D2: Arousal	2 (9)	-	0:12	-
D3: Frequency of Sexual Activity	1 (8)	-	0:12	Tally count of sexual activities
D4: Receptivity/Initiation	3	-	0:15	-
D5: Problems Affecting Sexual Function	2 (9)	-	0:12	-
D6: Relationship satisfaction	3	-	0:12	-
D7: Problems Affecting Sexual Function	4 (15)	-	0:16	-
Sexual Inhibition Scale/Sexual Excitation Scale (SIS/SES)	45	0.84	-	Individual differences in sexual activities
<i>Subscales</i>				
Excitation	20	25.08	0:80	-
Inhibition Due to the Threat of Performance Failure	14	6.67	0:56	-
Inhibition Due to the Threat of Performance Consequences	11	0.77	0:44	-
Mood				
Evaluation of Risks Scale (EVAR)	24	0.70	0:100	General propensity for risky behavior
Beck Depression Inventory-II (BDI)	21	0.88	0:63	Attitudes and symptoms of depression
Positive and Negative Affect Schedule (PANAS)	20	0.77	10:50	Positive and negative affect

correspondence. In women, hormonal fluctuations, such as those resulting from the menstrual cycle, may also contribute to impulsive overall decision behavior, specifically impulsive sexual decision-making, so the menstrual phase was included as a control variable to model any nuisance variability (Rupp et al. 2009). Determination of the menstrual phase for test scheduling was done using a counting method from the first day of prior menses and verified by later hormone assay from urine samples. Testing for the ovulatory phase session occurred between days 10 and 14 after the women report menstruation began and testing for the luteal phase occurred days 19–23 following menstruation.

Before each fMRI session, participants reported their recent alcohol and drug use for the last week and provided a small urine sample (20 mL) for later hormone assay. This urine sample was also used for a drug screen and to verify that participants were not pregnant. The urine samples remained in the refrigerator for the remainder of the session at which point they were transferred to deep freeze storage (-20°C). Samples were assayed for estradiol, testosterone, and progesterone measurement and to verify the phase of the menstrual cycle at the time of testing (Israel et al. 1972). Following the urine sample, if the drug screen and pregnancy tests were negative, participants moved on to the imaging step.

Imaging took place at the Indiana University Imaging Research Facility. Participants completed a practice run of the task outside of the scanner to familiarize them with task procedures. The procedure was conducted with a script programmed in Matlab 7.6 and the Psychophysics Toolbox (<https://www.mathworks.com>; <http://www.psychtoolbox.org>) (Brainard 1997; Pelli 1997) on an Apple MacBook Pro laptop. The practice run was a shortened version of the actual data collection runs and used a different set of pictures from all of the same cue categories as were used in the actual study task. After participants understood the task, they were comfortably positioned in the MRI scanner (3T Siemens TRIO).

Functional scanning of 280 total trials was broken up into five ~ 7-min runs. The protocol for each run was based on a rapid event-related design with 56 trials all separated by variable-length inter-trial intervals. Each interval was 2, 4, or 6 s long and the different length intervals were used in a ratio of 4:2:1, respectively. On each trial, a stimulus from one of the four cue categories was pseudorandomly chosen without replacement, such that 14 cues from each category were presented during each run, 7 with low-risk information, and 7 with high-risk information. Across the five runs, this protocol produced 35 trials for each of the eight combinations of cue category (4) and risk condition (2). In the current article, only

Table 2 Participant demographics, SSAGA problem counts, substance use, and mood ratings

	Control (<i>n</i> =16)	Alcohol dependent (<i>n</i> =16)	Sig.
Age (years)	20.25 (1.57)	21.20 (2.08)	n.s. ^a
Education [<i>n</i> (%)]			
High school graduate	1 (%)	3 (%)	n.s. ^b
Some college	13 (%)	9 (%)	
College graduate	2 (%)	3 (%)	
SSAGA problem counts [mean (SD)]			
Alcohol problems	0.94 (1.34)	7.87 (3.07)	<.001 ^a
Cannabis problems	0.00 (0.00)	1.67 (3.37)	n.s. ^a
Recent substance use [mean (SD)]			
Alcohol frequency (days/weeks)	1.50 (1.21)	4.20 (1.15)	< .001 ^a
Alcohol quantity (drinks/week)	4.47 (4.62)	36.57 (18.10)	< .001 ^a
Cannabis frequency (days/week)	0.00 (0.00)	1.60 (2.44)	.014 ^a
Mood [mean (SD)]			
PANAS negative affect	12.19 (3.31)	13.50 (4.83)	n.s. ^a
PANAS positive affect	24.44 (7.74)	25.08 (6.97)	n.s. ^a
BDI	7.94 (9.59)	6.67 (5.88)	n.s. ^a
Sexuality [mean (SD)]			
<i>n</i> (%) currently in a sexual relationship	5 (%)	9 (%)	n.s. ^b
No. of different sexual partners (lifetime)	1.81 (2.10)	5.67 (3.92)	.003 ^a
No. of different sexual partners without condoms (past years)	0.25 (0.58)	1.07 (1.10)	.014 ^a
SOI	40.38 (17.94)	61.87 (29.61)	.020 ^a
BISF-W composite score	23.71 (14.45)	28.60 (15.11)	n.s. ^a
Sexual Experiences Survey	1.00 (2.00)	2.07 (1.98)	n.s. ^a
SES	49.69 (6.62)	49.00 (7.00)	n.s. ^a
SIS1-Performance failure	39.19 (4.48)	40.00 (5.34)	n.s. ^a
SIS2-Performance consequences	19.31 (3.77)	20.67 (4.88)	n.s. ^a
Risk taking [mean (SD)]			
EVAR	12.73 (1.98)	14.03 (2.54)	n.s. ^a

SSAGA Semi-structured assessment for the genetics of alcoholism; PANAS Positive and negative affect schedule; BDI Beck depression inventory; SOI Sociosexuality orientation inventory; SIS Sexual inhibition scale; SES Sexual excitation scale; EVAR Evaluation of risks scale

^a t-test; ^b Chi-square test

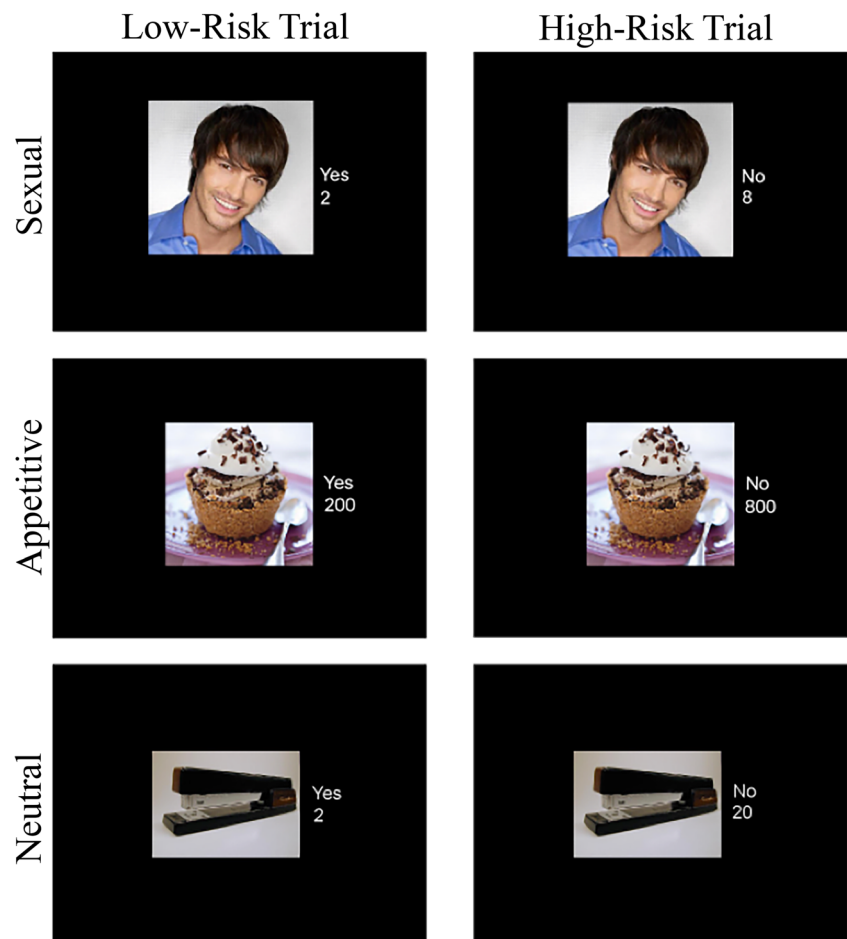
three cue categories were of interest (sexual, appetitive, and neutral), with the fourth (alcohol) only being used in statistical models as a regressor of no interest.

On each trial, the cue image was presented simultaneously with risk text for 4 s. During this time, participants appraised the combination of cue and risk information and rated their likelihood, using a button response pad, to have sex, eat food, or buy the item (or drink alcohol) on a four-point scale where 1 = very unlikely, 2 = unlikely, 3 = likely, and 4 = very likely (Fig. 1). The risk information conveyed the different situational context depending on the type of cue: for sexual cues, whether or not the male face shown usually used condoms and the number of sexual partners (low, $M \pm SD = 2 \pm 1$; high, 8 ± 1); for appetitive (food) cues, whether or not the food

establishment passed its latest health and safety inspection and the caloric content (low, $M \pm SD = 200 \pm 10$; high, 800 ± 10); for neutral (item) cues, whether or not the store had a return policy and the cost in dollars (low, $M \pm SD = 2 \pm 1$; high, 20 ± 1); and for alcohol cues (not included in the current analyses), whether or not the participant had a designated driver and how many alcohol units (1 unit = alcohol content in 1 shot, 1 glass of wine, or 1 beer depending on whether the alcohol cue depicted a cocktail, glass of wine, or beer) the drink contained (low, $M \pm SD = 1 \pm 1$; high, 6 ± 1). Specific risk number values were selected randomly on each trial, with a minimum value of 0 and no maximum value.

After functional imaging was completed, an anatomical scan was collected while the participant relaxed.

Fig. 1 Sample stimuli and risk scenarios. Each panel is a single trial. The word “yes” indicated low risk and “no” indicated high risk. In sexual decisions, the number indicated the number of total sexual partners and the yes/no answered whether or not the individual had an STD. In food decisions, the number referred to caloric content and the yes/no answered whether the food had passed a health safety inspection. In item decisions, the number referred to price and the yes/no answered whether or not the store had a return policy



MRI analysis procedures and parameters

Imaging parameters Imaging was carried out using a Siemens Magnetom Trio 3-T, whole-body MRI, and collected on a 32-channel phased-array head coil. Each fMRI session took about an hour, during which the following scans were acquired: (1) three-plane scout used for choosing slice planes for the remaining scans (10 s), (2) Gradient-echo T2* echo-planar imaging (EPI) scans for blood-oxygen-level-dependent (BOLD)-based functional neuroimaging (duration ~ 7 min, five scans/session, ~ 35 min total functional scanning), and (3) T1 3-D turbo-flash structural scan of the entire brain at high resolution (1-mm isotropic voxels) (~ 5 min). The functional pulse sequence had the following EPI parameters: echo time (TE) = 30 ms, flip angle = 70°, field of view = 240 × 240 mm, matrix 96 × 96, in-plane resolution = 2.5 mm slice thickness = 3.5 mm, gap thickness = 0 mm. A functional volume was 32 EPI slices acquired at a time of 62.5 ms per slice for a total volume acquisition time of 2 s [repetition time (TR) = 2]. Slices were acquired approximately parallel to the anterior commissure/posterior commissure (AC-PC) plane to efficiently cover the

entire brain. High-resolution T1-weighted anatomical volumes were acquired using Turbo-flash 3-D (TI = 900 ms, TE = 2.67 ms, TR = 1800 ms, flip angle = 9°) with 160 sagittal slices with a thickness of 1 mm and a field of view of 224 × 256 (voxel size = 1 × 1 × 1 mm).

Whole-brain GLM Imaging data were analyzed using FSL v4.1.9 (FMRIB Software Library; online at <http://www.fmrib.ox.ac.uk/fsl>, August 2012). GLM-based analysis in FSL was carried out with the fMRI Expert Analysis Tool (FEAT) (Jenkinson et al. 2012; Woolrich et al. 2009). Functional scans were co-registered to the MNI template (MNI-152 average brain). Functional scans were preprocessed using MCFLIRT for motion correction, the brain extraction tool (BET) for skull stripping, with a spatial smoothing FWHM window of 5 mm, and a high-pass temporal filter (Smith et al. 2004).

BOLD fMRI data were analyzed in a 4 × 2 × 2 × 2 full-factorial, whole-brain GLM analysis with decision type (male face/sexual, alcohol, food/appetitive, item/neutral), risk (high, low), and phase (follicular, luteal) as within-subject factors and group (control, AUD) as a between-subject factor. In addition, reaction time was included as a covariate for each run

for each participant (Grinband et al. 2008). The reaction time covariate was calculated separately for each first-level contrast by applying the same contrast to the mean reaction time across conditions. Before entry into the model, reaction time covariates were demeaned. Direct comparisons of alcohol decisions with control decisions (food and item) were documented in a previous article (Arcurio et al. 2015). Thus, the analyses focused on comparisons of sex decisions with control decisions. Thus, only three stimulus types were of interest (sexual, appetitive, neutral) and the fourth stimulus category (alcohol) was only included in the model to account for the variance attributable to those trials but which were not further analyzed. Likewise, the menstrual phase was not a variable of interest but was included as a factor in the model to account for any variance that may be attributable to testing on the different days.

The first-level analysis used custom predictors based on the timing protocol of each of the eight combinations of cue category (4) and risk information (2), convolved with a two-gamma hemodynamic response function. Outputs from the first-level analysis were contrasts among the sex, food, and item cues, combined with high- and low-risk conditions. In particular, we were interested in the first-level two-way contrast of $[(2 \times \text{SEX}) - (\text{APPETITIVE} + \text{NEUTRAL})]$ by (high risk – low risk).

The second-level analysis combined first-level outputs from separate runs for each level of the menstrual cycle phase factor for each participant. Effects of phase were negligible, making the contrast (luteal + follicular) the only relevant second-level output.

The third-level analysis combined second-level outputs across participants within each group (controls and AUD). Outputs from the third-level analysis were contrasts representing each group, both groups combined, and the difference between groups. The higher-level analyses were performed using a mixed-effects model (FLAME 1). The multiple testing problem was addressed by using a voxel-wise $z > 2.3$ threshold, which was then corrected at the cluster level with $\alpha = 0.05$ using random field theory (Worsley 2001).

ROI analysis Post-hoc region of interest analyses was conducted using the Harvard Oxford Cortical and Subcortical atlases to explore the specific patterns of activation across conditions in regions that were considered theoretically important to decision-making in AUD women (Makris et al. 2006; Frazier et al. 2005; Desikan et al. 2006) and that showed reliable activation in one of two whole-brain maps. ROI maps were created with fslmaths by extracting voxels that were both within the Harvard Oxford ROI map and showed reliable BOLD signal differences in the whole-brain GLM contrast map. The caudate and the putamen were localized from the map produced by the contrast of $((\text{SEX}^{\text{high risk}} - \text{SEX}^{\text{low risk}}) - (\text{APPETITIVE}^{\text{high risk}} - \text{APPETITIVE}^{\text{low risk}}))$. The mPFC

and the frontal pole were localized from the map produced by the contrast $(2 \times (\text{SEX}^{\text{high risk}} - \text{SEX}^{\text{low risk}}) - [(\text{APPETITIVE}^{\text{high risk}} - \text{APPETITIVE}^{\text{low risk}}) + (\text{NEUTRAL}^{\text{high risk}} - \text{NEUTRAL}^{\text{low risk}})])$. Initially, six ROIs were inspected, but only four are shown, and subsequent multiple comparison corrections reflect that number.

The beta-weights of each voxel in the resulting maps were extracted for each ROI. These beta-weight values were then placed into separate $2 \times 3 \times 2$ ANOVA analyses in SPSS with risk (high, low) and decision type (sexual, appetitive, neutral) as within-subject factors and group (controls, AUD) as a between-subject factor. As with all previous analyses, only three stimulus types were of interest, and the data were collapsed across levels of the menstrual cycle phase.

Behavioral task analysis

Endorsement ratings were measured on a 4-point Likert scale. Participants would use this scale to indicate how likely they were to have sex, drink, eat, or buy the image dependent on whether it was a sexual, alcohol, appetitive, or neutral control decision respectively. For each trial, the 4-point scale was binarized with very unlikely and unlikely responses coded as 0 and very likely and likely responses coded as 1. The mean across trials for each condition produced the endorsement likelihood dependent measure. Reaction times (RT) were also recorded for planned analysis and as a covariate in the fMRI GLM analysis, and the mean across trials per condition produced the dependent measure of RT.

Results

Sexuality

Mean (SD) scores for the control and AUD groups on the mood, sexuality, and risk-taking measures are provided in Table 2. Drinking frequency was positively correlated across both AUD and control women with measures of sexuality, sexual activity, and sexual risk taking (see Table 3, Figure OR1). In general, women with AUD engaged in more risky sexual activities, had more sexual arousal, had more sexual partners, and were more likely to engage in short-term sexual encounters than women in the control group (Table 2).

Women in the AUD group reported more lifetime sexual partners than control women ($t(29) = 3.760, p = 0.001$) and more unprotected sexual encounters (number of partners without condoms) over the past year than did control women ($t(29) = -2.810, p = 0.008$). In addition, women with AUD scored higher than control women on the SOI (see Tables 1, 2, and SI), indicating a greater tendency to engage in short-term sexual encounters ($t(29) = -2.787, p = 0.009$). On the

Table 3 Correlation of sexuality measures and drinking frequencies

	<i>r</i>	Sig.
Number of sexual partners	$r_s = 0.608$	$p < 0.002$
Number of partners with condoms	$r_s = 0.399$	$p = 0.046$
Number of partners without condoms	$r_s = 0.443$	$p = 0.031$
SOI	$r(29) = 0.443$	$p = 0.044$
BISF-W D1	$r(29) = 0.219$	n.s.
BISF-W D2	$r(29) = 0.388$	$p = 0.054$
BISF-W D3	$r_s = 0.440$	$p = 0.031$
BISF-W D4	$r(29) = 0.279$	n.s.
BISF-W D5	$r(29) = 0.131$	n.s.
BISF-W D6	$r(29) = 0.010$	n.s.
BISF-W D7	$r(29) = 0.036$	n.s.

False discovery rate (FDR) corrected p -values reported

BISF-W (see Tables 1 and 2, Fig OR1), women with AUD scored higher on sexual frequency than control women ($t(29) = -2.306$, $p = 0.028$).

Correlations between alcohol use measures and sexuality measures were computed using either Pearson's r or Spearman's ρ in correspondence with the statistical nature of the response variable (Hauke and Kossowski 2011). The number of lifetime sexual partners was positively correlated with the frequency of alcohol use across all individuals in both groups ($r_s = 0.608$, $p < 0.001$) and the number of unprotected sexual encounters was positively correlated with drinking frequency ($r_s(29) = 0.443$, $p = 0.031$). Also, the number of partners with whom participants did use condoms was negatively correlated with drinking frequency ($r_s(29) = -0.399$, $p = 0.046$). In addition, SOI scores were positively correlated with drinking frequency ($r(29) = 0.443$, $p = 0.044$). Finally, the BISF-W sexual frequency measure was positively correlated with drinking frequency ($r_s(29) = 0.440$, $p = 0.031$) and the BISF-W sexual arousal measure was positively correlated with the amount of alcohol consumption ($r(29) = 0.39$, $p = .054$).

All correlation p -values reported were corrected using the false discovery rate to account for multiple comparisons. There were no significant group differences on the PANAS, BDI, SIS/SES, Sexual Experiences Survey, EVAR, or BISF-W dimensions 1, 2, 4, and 5 ($p > 0.05$; see Tables 1 and 2).

Decision-making task behavior

Endorsement likelihood

A repeated-measures ANOVA with endorsement likelihood of sexual decisions as the dependent measure and phase, risk, and group as the independent variables showed only a main effect of risk, with low-risk scenarios endorsed more ($M = .539$, $SE = .008$) than high-risk scenarios ($M = .062$, $SE = .003$, $F(1, 29) = 98.322$, $p < .001$, $\eta_p^2 = .772$).

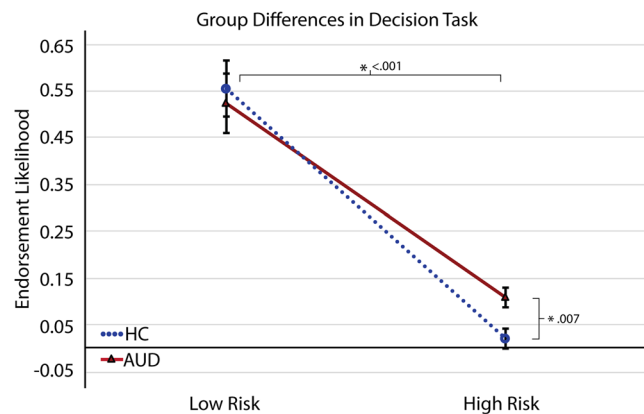


Fig. 2 Significant behavioral effects on endorsement likelihood or the self-reported probability of participation in sexual activity. It can clearly be seen that risky sexual decision-making is uniquely affected for the AUD population, shown by less reduction in endorsement likelihood. The main effect of risk is also apparent, with endorsements decreasing significantly for high-risk decisions

Following previous research (Arcurio et al. 2015; Nikoulina et al. 2020) where effects of other factors were largely confined to the high-risk condition, a planned comparison found that AUD women reliably endorsed risky sexual activity ($M = 0.108$, $SE = .031$) more than the control group ($M = .019$, $SE = .006$, $t(29) = -2.88$, $p = .007$) (Fig. 2).

Reaction times (RTs)

The mean reaction time for sex decisions for all groups and conditions was 1.76 s ($SE = .013$). A repeated-measures ANOVA with reaction time as the dependent measure and phase, risk, and group as the independent variables found no significant effects.

Whole-brain GLM

Risky sexual decisions

To explore the effect of risk on sexual decisions, we specifically looked at the first-level contrast ($SEX^{high\ risk} - SEX^{low\ risk}$) for the positive and negative tails of three separate third-level contrasts: controls alone, AUDs alone, and AUDs minus controls.

In controls, while making high- compared to low-risk sexual decisions, there was widespread deactivation of key reward, visual, and frontal regions (Fig. 3a). Reward regions included the caudate, putamen, substantia nigra, and VTA indicating that increasing risk decreased activation of limbic regions. Frontal regions that deactivated in response to high risk were part of the central executive network and included the MFG, the inferior frontal gyrus (IFG), the frontal insular cortex (FIC), and the medial prefrontal cortex (mPFC). Core regions of the salience network (the dACC and the AIC) also

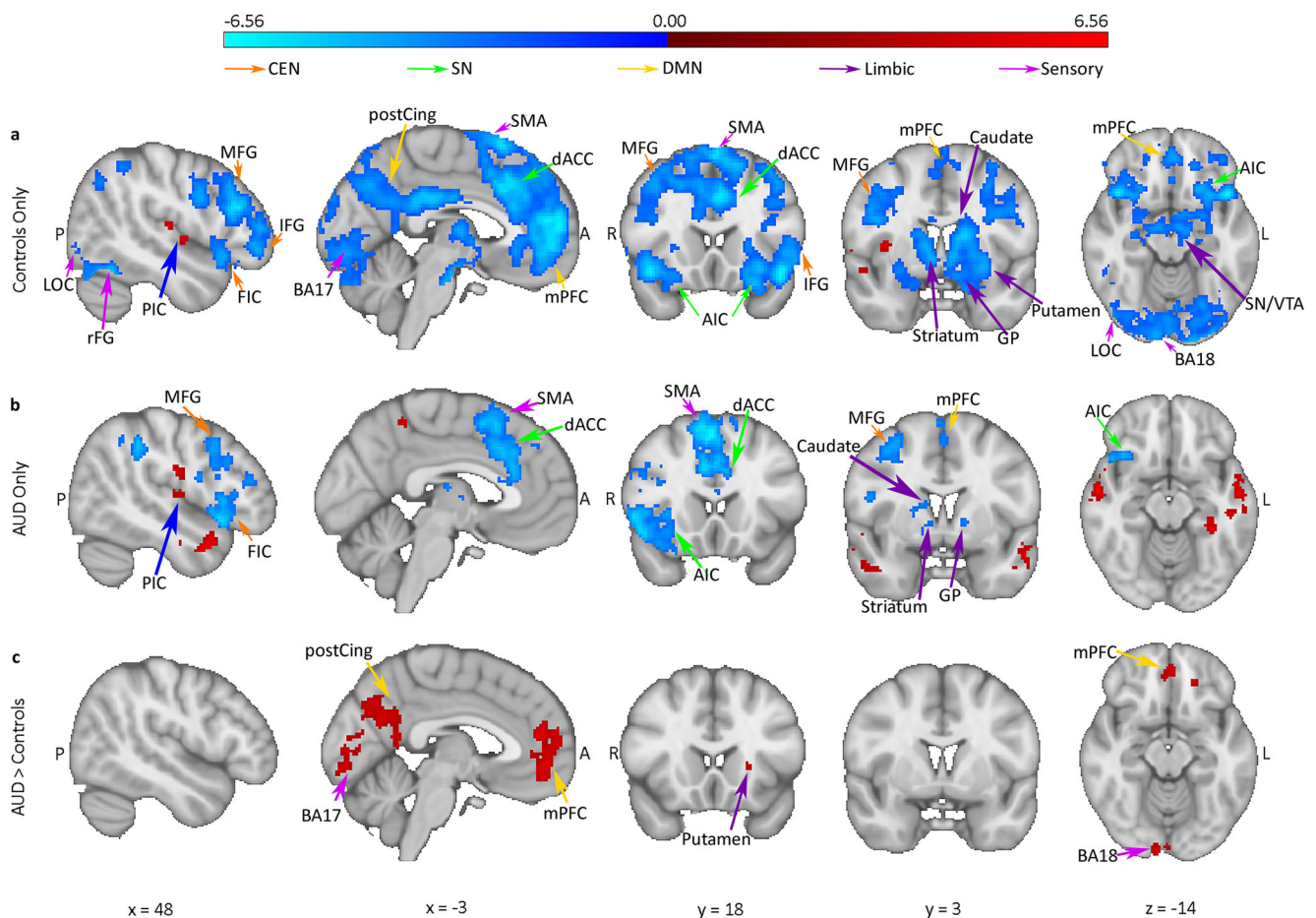


Fig. 3 Statistical map for high-risk sexual decisions minus low-risk sexual decisions. Specifically, the contrast ($SEX^{high\ risk} - SEX^{low\ risk}$). Bar along the top indicates z-threshold values. **a** Importantly, controls show widespread deactivation of limbic/reward and sensory regions in addition

to key regions of the CEN, DMN, and SN. **b** AUDs show a similar pattern but fail to deactivate DMN regions, some CEN regions, and some SN regions (e.g., left AIC). **c** The difference map of AUD-Controls

showed deactivation in response to increased risk in controls. Visual regions such as the early visual cortex (BA17,18) and the fusiform gyrus also showed deactivation in response to high risk in controls. The supplemental motor area (SMA) also showed deactivation, indicating that **inhibition of motor planning** may be a part of sexual decision-making in controls individuals.

In controls, there was very little increased activity with high- over low-risk decisions, with the only significant region found in the posterior insular cortex (PIC).

In AUD individuals, the pattern was similar to controls but greatly attenuated (Fig. 3b). The spatial extent of the deactivation of key reward, visual, salience, and executive control regions appeared to be overall less than controls. In addition, AUD individuals showed increased bilateral activity in superior and middle temporal gyri with high-risk decisions.

Contrasting AUDs to controls (Fig. 3c) highlighted the areas of attenuated deactivation in AUDs. The AUDs appeared to **show hyperactivation in frontal, reward, visual regions, and DMN regions**. However, what is clear from the

individual group maps is that the apparent hyperactivation is actually caused by a lack of deactivation in the AUD group for the high- compared to low-risk condition. Importantly, hyperactivation in AUDs has been previously reported by the literature for both alcohol and sexual stimuli (Yalachkov et al. 2010; Robinson and Berridge 2008; Chase et al. 2011; Noori et al. 2016).

Sexual vs. appetitive/neutral decisions

We next investigated the difference between sexual decisions versus appetitive and neutral decisions (control decisions), looking only at the baseline low-risk condition. The specific first-level contrast used was $(2 \times SEX^{low\ risk} - (APPETITIVE^{low\ risk} + NEUTRAL^{low\ risk}))$ and this was again evaluated for the positive and negative tails of three separate third-level contrasts: controls alone, AUDs alone, and AUDs minus controls.

In controls (Fig. 4a), the general pattern included activation of limbic/reward regions, visual regions, and decision regions.

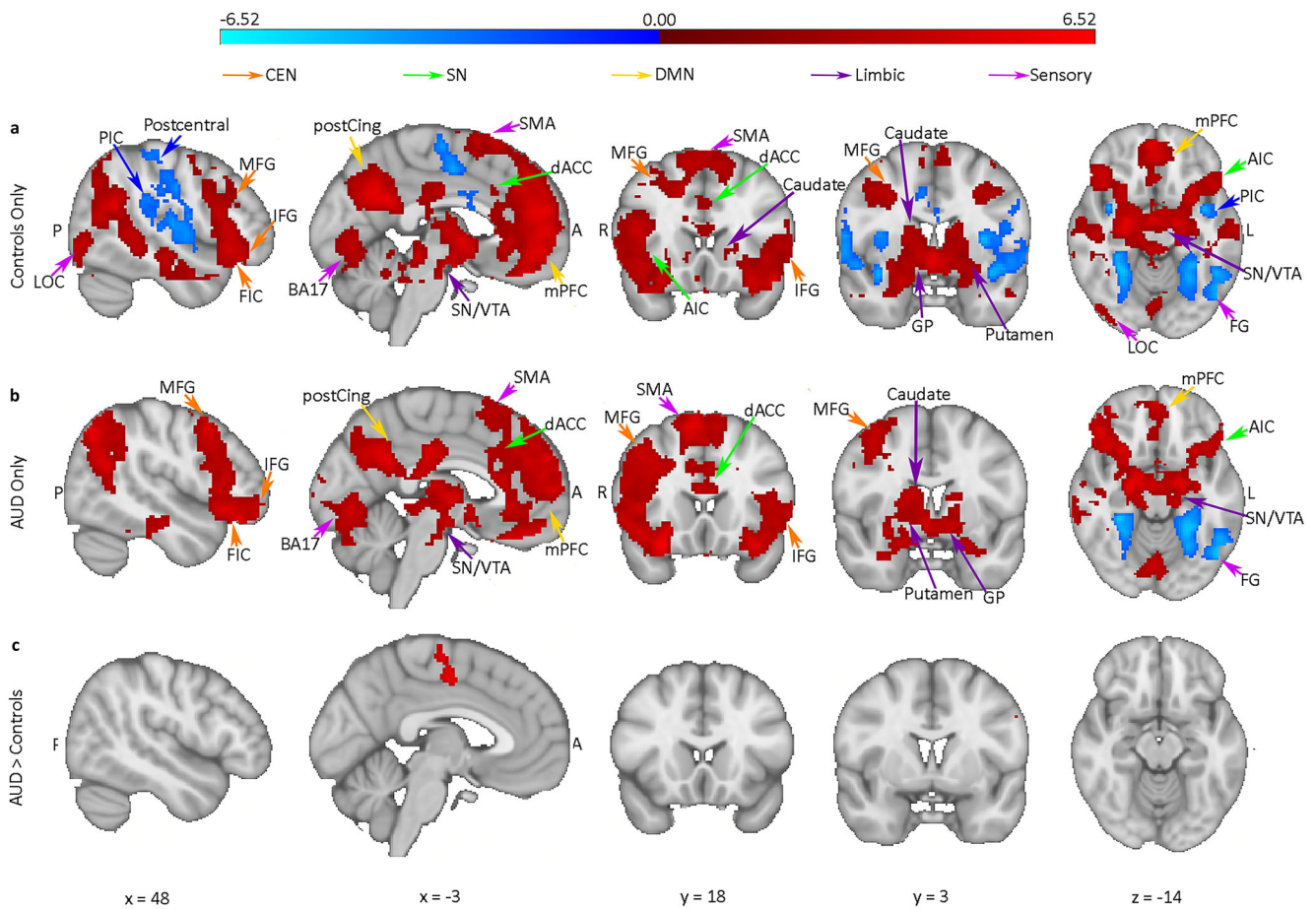


Fig. 4 Statistical map for low-risk sexual decisions minus low-risk food and item decisions. Specifically, the contrast $(SEX^{low\ risk} - (FOOD^{low\ risk} + ITEM^{low\ risk}))$. Bar along the top indicates z-threshold values. In low risk, controls show widespread activation of key reward/limbic, visual,

and decision-making areas. Specifically, there is activation of the CEN, DMN, and SN. Some deactivation of the left fusiform is observed. AUDs show a similar but attenuated pattern of activation. **c** The difference map of AUD-controls

Decision regions such as the MFG, IFG, and FIC represent activation of CEN, whereas regions such as the mPFC represent DMN activation. There was also activation of the core SN (the dACC and the AIC). Visual regions such as the LOC and early visual cortex (BA17) showed activation, while the left fusiform gyrus showed deactivation. The SMA also showed activation, indicating motor preparation. There was also deactivation of PIC, postcentral gyrus, and temporal lobe.

In AUD individuals (Fig. 4b), a similar pattern of activation to controls was observed. Comparing AUD individuals to controls (Fig. 4c), there were virtually no differences, indicating that it is likely in situations of high risk where AUD individuals are most vulnerable.

Risky sexual vs. appetitive/neutral decisions

Finally, we investigated the more complex interactions between risk, group, and decision type using the specific first-level contrast $(2 \times (SEX^{high\ risk} - SEX^{low\ risk}) - [(APPETITIVE^{high\ risk} - APPETITIVE^{low\ risk}) + (NEUTRAL^{high\ risk} - NEUTRAL^{low\ risk})])$ and this was again

evaluated for the positive and negative tails of three separate third-level contrasts: controls alone, AUDs alone, and AUDs minus controls. Essentially, the examination of this 3-way factorial analysis allows the interpretation of the high-risk sexual decision condition in AUDs, compared to the baseline low-risk condition, compared to control decisions (appetitive and neutral), and compared to control individuals, simultaneously.

First, examining the 2-way interaction of decision type by risk in controls only (Fig. 5a), we again found significant deactivation in fronto-striatal systems and visual regions for high-risk sexual decisions compared to low-risk sexual decisions and compared to high-risk control decisions. We also found deactivation of the anterior insula which is thought to be a hub for regulating the salience, central executive, and default mode networks (Menon and Uddin 2010; Arcurio et al. 2015). Bilateral deactivation of reward/limbic systems was observed again, specifically in the putamen, caudate, globus pallidus, ventral tegmental area (VTA), and substantia nigra.

Second, examining the 2-way interaction of decision type by risk in the AUD group showed an attenuated pattern

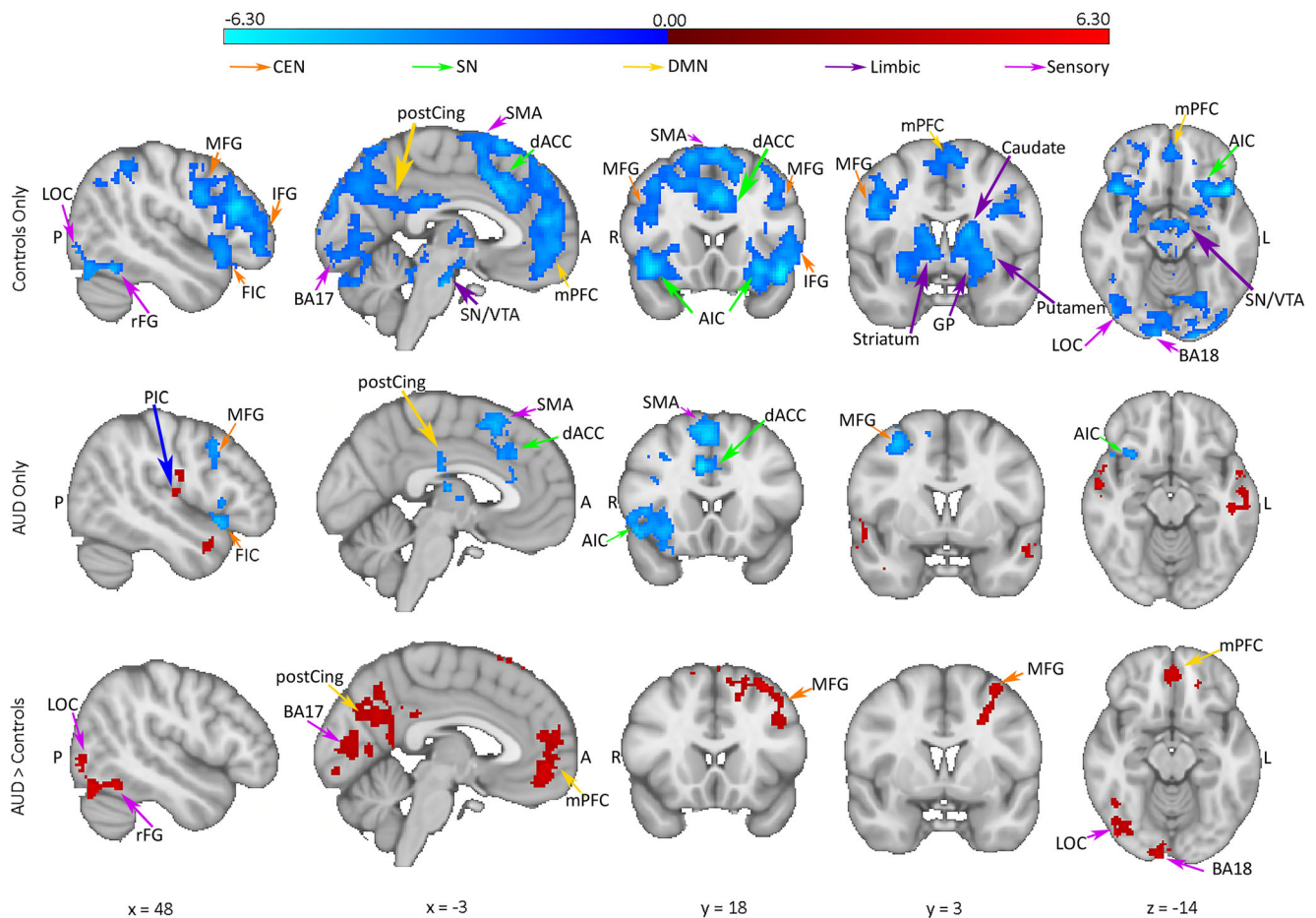


Fig. 5 Statistical map for sexual decisions minus food and item decisions. Specifically, the contrast $2 \times (\text{SEX}^{\text{high risk}} - \text{SEX}^{\text{low risk}}) - [(\text{FOOD}^{\text{high risk}} - \text{FOOD}^{\text{low risk}}) + (\text{ITEM}^{\text{high risk}} - \text{ITEM}^{\text{low risk}})]$. Bar along the top indicates z-threshold values. It is clear that controls deactivate regions

of the DMN, CEN, SN, reward regions, and visual regions, whereas AUDs show attenuated deactivation in visual, DMN, and CEN regions. **c** The difference map of AUD-controls

compared to controls (Fig. 5b). There was some deactivation of frontal areas and the right anterior insula. The AUD group showed some deactivation in SN in the left AIC and the dACC. That group also showed some deactivation of right MFG indicating highly attenuated CEN deactivation compared to controls. In contrast to controls, women with AUD showed increased activation in right posterior insula and bilateral superior temporal gyrus.

Direct comparison of the AUD and control groups via the three-way interaction of decision type by risk by group (Fig. 5c) revealed hyperactivation in visual and fronto-striatal regions. The direction of this effect (i.e., hyperactivation) must be interpreted with respect to all of the previous results. Specifically, many executive control, reward, and visual regions showed this heightened activity for high- versus low-risk sexual decisions compared to control decisions. However, the separate group maps show that hyperactivation in the AUD group was actually driven by a lack of deactivation; this deactivation was evident in controls for high- versus

low-risk sexual decisions compared to control decisions. Similar hyperactivations have been reported previously in the substance abuse literature, although our results suggest a new interpretation of these previous results: hyperactivation in AUD women may actually reflect a lack of deactivation normally seen in controls with high-risk decisions (Yalachkov et al. 2010; Robinson and Berridge 2008; Chase et al. 2011).

ROI analysis

A limited ROI analysis was performed to confirm that the pattern of activation across experimental factors was being interpreted correctly from the various statistical maps reported above. ROIs were derived from activation maps and the Harvard Oxford structural maps (see Methods and materials and Fig. 6a). Across all ROIs there was high consistency in direction and pattern of effects. A predominant pattern that emerged was the increased difference between high- and low-risk activation for controls compared to individuals with AUD only for sexual decisions. This difference

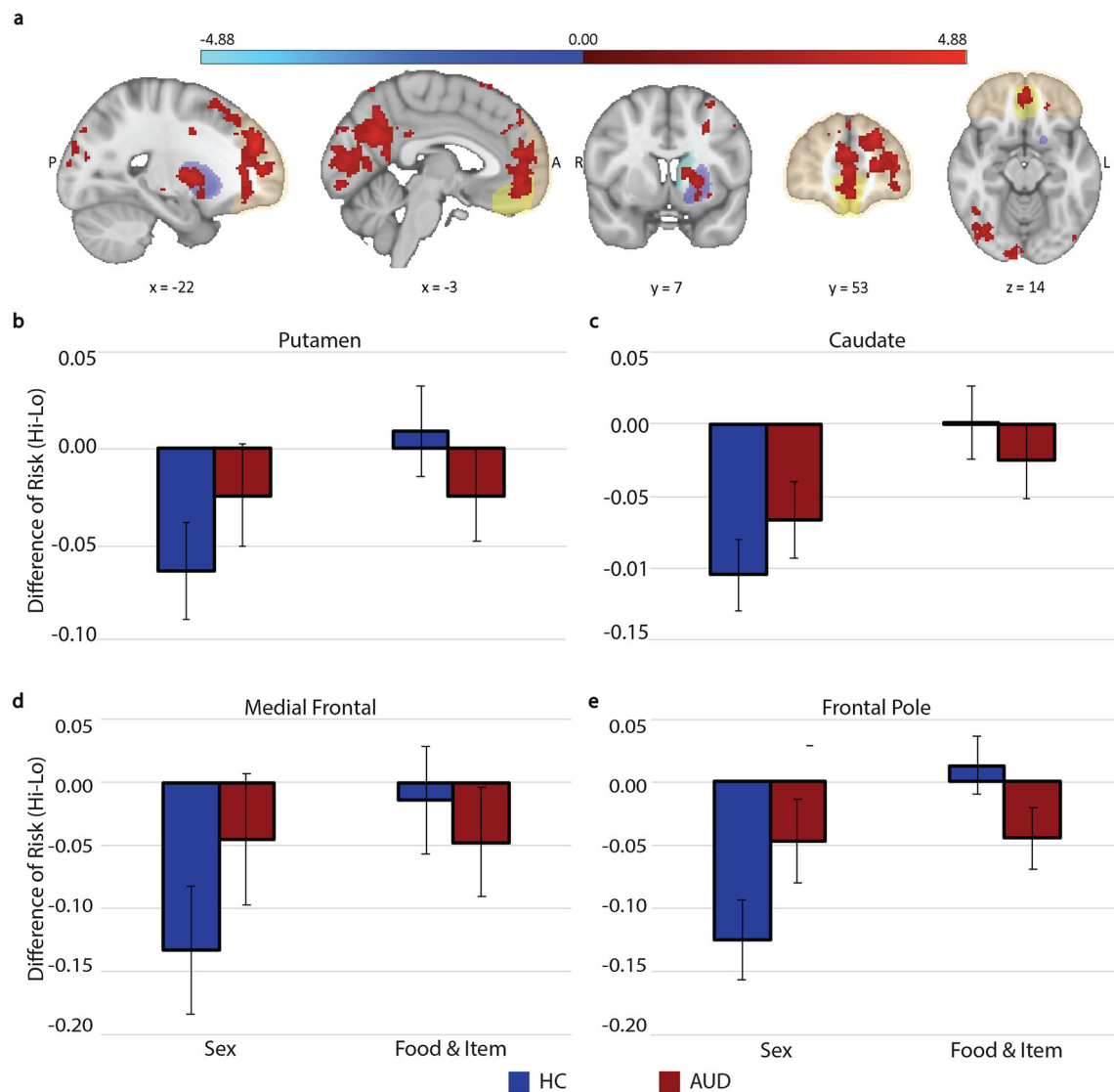


Fig. 6 Results of post-hoc ROI analysis, the difference of high-low risk for each condition shown in the bars. Red bars represent AUDs and blue represent controls. Standard error bars shown. **a** ROI regions. Shaded coloration used to identify Harvard Oxford ROIs, red indicates activations from AUDs > HCs contrast maps of sexual decisions. Blue shading represents left caudate, purple is left putamen, pale orange is the frontal pole, yellow is medial prefrontal. **b** In the putamen, an important reward processing region, AUD subjects fail to reduce activation in response to

high-risk sexual decisions whereas controls show no difficulty. **c** During high-risk sexual decisions, AUDs maintain high activation of the caudate, a region is known to process both substance and natural reward. **d** AUD subjects modulate their response to appetitive and neutral stimuli in the medial prefrontal cortex but fail to modulate their response to high-risk sexual decisions. **e** In this important decision region, it is found that participants have an increased response to sexual decisions and that AUD participants fail to modulate neural activation

indicates that in high-risk situations, controls deactivated the various brain regions more than AUD individuals for sexual decisions (Fig. 6). The Bonferroni corrected alpha for the multiple ROI analyses was set at the significance threshold of $p = 0.007$.

Executive control/decision-making ROIs

Patterns of activation were consistent across both executive control and decision-making ROIs. The medial prefrontal cortex of the DMN (Fig. 6d) showed a main effect of risk, indicating reduced activation during high-risk decisions compared to low-risk decisions for both healthy controls and AUD

participants ($F(1, 29) = 75.046, p < .001, \eta_p^2 = .721$). The medial prefrontal cortex also showed a main effect of decision type that indicated that sexual decisions resulted in greater activation than appetitive and neutral decisions ($F(1, 29) = 14.305, p < .001, \eta_p^2 = .330$). In the frontal pole (Fig. 6e), which included the activation of MFG making it part of the CEN, both the main effect of risk ($F(1, 29) = 58.057, p < .001, \eta_p^2 = 0.667$) and the main effect of decision type ($F(1, 29) = 30.202, p < .001, \eta_p^2 = .510$) indicated that lower risk and sexual decisions showed more activation compared to higher risk and control decisions for both AUD individuals and healthy controls.

Reward ROIs

The reward-related activity was more pronounced in the left caudate compared to the left putamen, but both activity patterns were largely consistent. In the left putamen (Fig. 6b), there was only a main effect of risk ($F(1, 29)=10.731$, $p=.003$, $\eta_p^2=.270$), indicating a similar pattern to the executive control ROIs where low risk produced greater activation than the high risk for both AUD individuals and controls. In the left caudate (Fig. 6c), the same main effect of risk emerged ($F(1,29)=23.956$, $p<.001$, $\eta_p^2=.452$). The left caudate also showed a main effect of decision type ($F(1, 29)=9.403$, $p=.008$, $\eta_p^2=.245$) that indicated higher activation for sexual compared to control decisions in both AUD individuals and controls. The left caudate also showed an interaction between risk and decision type that was driven by a decreased difference in activation between sexual and control decisions during high-risk conditions ($F(1, 29)=11.880$, $p=.002$, $\eta_p^2=.291$). Although no group differences were reliable, it seems this interaction was driven by the controls' ability to deactivate reward response for high-risk decisions.

Discussion

We combined self-report measures with a decision-making task and neuroimaging data to investigate sexual decision-making in women with and without AUD. Women with AUD made riskier hypothetical sexual decisions than control women. This behavioral difference was accompanied by hyperactivation in fronto-striatal and visual brain regions for high-risk sexual decisions in AUD women compared to control women. The pattern of results mimicked the results of a previous study of high-risk drinking decisions in women with AUD (Arcurio et al. 2015). Importantly, the results demonstrated that the observed hyperactivation in AUD women was driven by a lack of deactivation, which was shown to be the baseline response to high-risk sexual decisions in control women. That is, the critical result was that control women's less risky behavior appeared to be driven by deactivation of brain regions, whereas AUD women failed to deactivate those same brain responses, leading to riskier decisions. The findings were consistent across whole-brain mapping and ROI analysis approaches. Finally, the differences in activation between sexual and appetitive/neutral decisions indicated that high-risk sexual decisions recruited increased executive processing that AUD women failed to regulate as well as control women in response to heightened risk.

Women with AUD showed a greater likelihood to choose to engage in more hypothetical high-risk sexual activity than control women, whereas there was no similar group difference for low-risk scenarios. In addition, sexual frequency, riskiness, and enjoyment were positively correlated with the

frequency of drinking. This suggested that women with AUD may be most vulnerable to poor sexual decision-making in situations where poor decisions may place them at the greatest risk. In the present study, these differences in decision behavior were found only for sexual decisions, not for appetitive or neutral decisions, indicating that sexual decisions may be uniquely affected in women with AUD, that is, the disinhibition present for these sexual decisions does not seem to reflect general disinhibition. Importantly, the women completed this decision-making task while sober, suggesting that chronic alcohol use may produce lasting impairments to decision behavior irrespective of intoxication state. However, as the present data were not longitudinal in nature, additional research is needed to verify this conclusively.

High-risk sexual decisions showed hyperactivations in women with AUD in frontal, reward, and visual regions. Importantly, these clusters were found in key brain regions implicated in addiction and sexual decision-making. Combined with the observation of riskier sexual decision behavior, this may indicate dysfunction specifically in women with AUD, specifically for sexual decisions, and specifically in high-risk situations. To reiterate a point made earlier, hyperactivation of these regions in AUD women appears to actually reflect a failure to deactivate, as deactivation of these regions was the result of high-risk sexual decisions in control women. The global pattern of results is consistent with previous research. Chronic alcohol misuse has been shown to alter frontal, visual, and reward regions structurally in animal models and functionally in humans (Jentsch and Taylor 1999; Chase et al. 2011; Yalachkov et al. 2010; Hanlon et al. 2014). The frontal lobe has been cited in human studies as promoting sexual activity (Baird et al. 2007). The current results, combined with previous results, suggest that the strong co-occurrence of drinking and sexual behavior in women with AUD may sensitize frontal, reward, and visual brain regions for sexual cues in a similar way to alcohol cues. The results indicate that these brain regions may be important targets for reducing risky sexual behavior in this population.

Results in specific brain regions are also consistent with previous work. The mPFC has been shown to positively correlate with drinking frequency and has also shown experience-dependent hyperactivations to decision contexts (Grüsser et al. 2004; Goldstein and Volkow 2011). The mPFC is a part of the DMN, which indicates that deactivation of mPFC may improve decision-making (Menon and Uddin 2010). Additionally, lesion studies in female rats have demonstrated the importance of the mPFC for increasing sexual behaviors (Afonso et al. 2007). The frontal pole is a large region at the most anterior region of the PFC that is broadly known to be vital for executive function and decision-making (Goldstein and Volkow 2011; Jentsch and Taylor 1999). Limbic/reward regions (nucleus accumbens, caudate, putamen) and emotion regions (amygdala) project directly to the PFC and regulate

the learning of salient information. These regions are also implicated broadly in substance use disorders and addiction, suggesting that learning and decision-making regarding the salient drug-related information are affected by AUD.

Although the focus of the study was sexual decision-making and sexual cue reactivity, it is worthwhile noting that the pattern of results with appetitive (food) cues was also consistent with previous work. In particular, the pattern of activation in the caudate and putamen showed a pronounced BOLD signal decrease for high- versus low-risk appetitive decisions. This pattern mimics the classic pattern considered to be the anhedonic state of substance-dependent individuals whereby natural rewards (like food) produce less neural activity in substance-dependent brains (Koob and Le Moal 2005). The result is also consistent with the opponent-process theory, which asserts that people with AUD should be less motivated by other appetitive cues. Opponent process theory would predict that the women with AUD would show hypoactivation in frontal and reward regions with appetitive cues compared to controls (Blum et al. 2000). In fact, the current results did show hypoactivation in the reward regions of AUD women compared to control women for appetitive cues, suggesting that appetitive cues that do not co-occur with alcohol use become devalued with increased alcohol consumption (Koob and Le Moal 2005).

Another model of addiction is incentive-sensitization (Robinson and Berridge 1993, 2008). Opponent processes and incentive processes are likely regulated by different neural circuits (Robinson and Berridge 2008; Blum et al. 2000). One key factor in incentive-sensitization is the hyperactivation of neural responses when presented with substance cues. Importantly, sensitization can generalize to other cues and behaviors that are associated with or co-occur with substance use, making the co-occurrence of behaviors a second key factor in incentive-sensitization theory. The women with AUD in our sample may have more often paired alcohol use and sexual activity due to the increased prevalence of both activities for these women compared to controls. It is possible that the similarity between the hyperactivation for sexual cues seen in our sample and the hyperactivation seen for alcohol cues seen in previous work (Arcurio et al. 2015; Chase et al. 2011; Yalachkov et al. 2010) could be partially explained by incentive sensitization theory (Davis et al. 2007; Koob and Le Moal 2005; Robinson and Berridge 2008). In animal models and in Parkinson's patients, increased dopamine release has been associated with sexual activity (Kohlert and Meisel 1999; Graham and Pfaus 2010; Giladi et al. 2007) and can result in motor preparation for behaviors previously paired with alcohol use (Franken 2003; Kalivas and O'Brien 2008), behaviors which could include sexual activity (Akins et al. 2017; Gill et al. 2015; Bolin 2012; Akins and Geary 2008). Research in quails and rodents has specifically shown that incentive-sensitization can generalize to sexual cues if drug reward

and sexual activity are paired (Akins et al. 2017; Gill et al. 2015; Bolin 2012; Akins and Geary 2008; Bradley and Meisel 2001; Graham and Pfaus 2010; Mitchell and Stewart 1990a, 1990b). Although speculative, the results reported are consistent with incentive-sensitization of alcohol and sexual cues in women with AUD and maybe the first step in extending the established animal research to humans (Kohlert and Meisel 1999; Graham and Pfaus 2010; Giladi et al. 2007).

Limitations

The present study had several significant strengths, including comprehensive clinical assessments of alcohol use history and current AUD, urine sampling to ensure sobriety at testing, and brain imaging during ecological decision-making. However, there were some limitations of note. The sample size of the current study was adequate for the effects reported, but modest, with only 15 individuals with AUD and 16 control participants. The sample size would not have allowed for the detection of menstrual phase-related influences that would surely have a much smaller effect size. As such, phase was included in the statistical models only as a factor of no interest. In addition, although hormone data were collected, correlating those outcomes with BOLD measures would have required a larger sample size. Thus, the hormone data were only used to confirm the menstrual cycle phase. In addition, while our study examined a carefully selected sample of young adult AUD women, non-AUD heavy binge drinking women and/or older women were excluded. Therefore, the results may not generalize to older women, women who drink heavily, but do not meet the criteria for AUD, or men (Truman et al. 2013; Tan et al. 2015). Similarly, our sample specifically excluded those with comorbid mental health disorders, current illicit drug use, and non-alcohol substance use disorders, so the degree to which the present results may generalize to those groups is unclear. Last, the longitudinal effects of sexual behavior need to be investigated to better understand the appropriate theoretical implications of the present results. Future research could combine the fMRI methods described here with a longitudinal ecological momentary assessment of real-world drinking behavior and risky sexual activity to evaluate whether sexual behavior is incentivized alongside alcohol using the same mechanisms.

Conclusions

The differential neural activation in decision-making, reward, and visual regions we observed for women with AUD showed that hyperactivations during sexual

decisions were driven by failures to deactivate key regions of the SN, CEN, and DMN. We had expected to see hyperactivation in BOLD response based on the previous literature, and our results were consistent with this. However, our design allowed us to examine the activation patterns more carefully and elaborated upon the cause of the hyperactivations. This led to our conclusion that AUD women were not actually increasing activation for sexual decisions but instead were failing to deactivate key regions to the same extent as healthy control women. The present data suggest that women with AUD are particularly vulnerable to risky sexual decision-making, and highlight the utility and importance of incorporating information regarding alcohol and sexual decision-making into education and treatment approaches with this population.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00213-021-05815-w>.

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Author contributions K.L. Folco wrote the manuscript and contributed to portions of data analysis, L.R. Arcurio and T.W. James designed experiments, collected and analyzed data, and contributed to portions of the manuscript text. D.J. Fridberg contributed to preliminary analyses and manuscript preparation. P.R. Finn managed recruiting, screening, and testing of participants. J.R. Heiman provided guidance, expertise, and insight regarding sexual behavior to experimental design, analysis, and manuscript revisions.

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Declarations

Conflict of interest The authors declare no conflict of interest.

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